

Wilson Huang

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WORK EXPERIENCE

Software Engineer

Feb. 2025 – Present

LinkedIn

Sunnyvale, CA

- Designed, built, and own a REST API letting enterprise integration partners pull HR events on demand, built end-to-end across 5 repositories (FastAPI, SQLAlchemy, Azure SQL, Azure APIM, React, Airflow); shipped zero-to-production in 3 months with 88% test coverage, replacing manual validation and projected to cut quarterly reconciliation errors ~69%.
- Designed authorization letting one API client hold multiple event subscriptions, each granting access to different data columns: queries filter per subscription and merge results (`UNION ALL`) into a single stable, paginated response, with zero breaking changes for existing clients.
- Built an event-driven approval pipeline: employee time-off and time-entry events land in Azure Event Hub, and a Function App routes approval requests to managers via Teams/Slack bots, backed by an internal Workday API that approves or denies entries by Workday ID.
- Guaranteed at-least-once event delivery via a transactional outbox (Postgres `FOR UPDATE SKIP LOCKED`, Airflow reconciliation sweep) and secured the pipeline so only the employee's manager can approve, verifying responder identity against Workday.
- Root-caused and resolved production incidents, including intermittent database connection failures traced to pyodbc's driver-level pool caching connections past Azure Managed Identity token expiry, and a silent API gateway bug where gzip-compressed responses bypassed APIM's response-rewrite policy and leaked internal hostnames.
- Led zero-downtime migrations: Python 3.12 across 9 backend microservices and team infrastructure to Azure Linux 3 (VMs, load balancers, Redis) via staged API gateway cutovers, decommissioning legacy database and Airflow servers for ~\$13–15K/year in savings.
- Technical interviewer for LinkedIn's REACH apprenticeship (5 technical interviews, 20+ screenings); mentor incoming apprentices and onboarded an external contractor onto the integration portal codebase.

Full-Stack Software Engineering Training (Self-Directed)

May 2023 – Jan. 2025

The Odin Project | DeepLearning.AI & Stanford Online

- Completed The Odin Project's full-stack curriculum plus Andrew Ng's Machine Learning and Deep Learning Specializations; built and deployed The OdinBook, a full-stack social platform with JWT authentication, dynamic feeds, image uploads, and infinite scroll (React, Node.js, Express, MongoDB).

Undergraduate Research Assistant

May 2022 – May 2023

Human Neuromechanics Laboratory, University of Florida

Gainesville, FL

- Developed MATLAB pipeline converting IMU accelerometer data to velocity metrics for gait analysis in vision-impaired populations, validating results across 20+ test subjects.

PROJECTS

Senior Design: ElectroLarynx Speech Enhancement | *Python, TensorFlow, Swift*

- Led 4-person team building a prototype iOS app using a custom TensorFlow model converting electrolarynx speech to natural voice for laryngectomy patients; achieved 95% detection accuracy through iterative optimization on 200+ audio samples.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, Swift, MATLAB

Backend & APIs: FastAPI, Flask, SQLAlchemy, Pydantic, REST APIs, OpenAPI, OAuth 2.0, Node.js, React

Cloud & Infrastructure: Azure (APIM, Event Hubs, Functions, Key Vault, Managed Identity, Azure SQL), Apache Airflow, Redis, Docker, GitHub Actions, CI/CD, Linux, MSSQL, MySQL, MongoDB

Practices: System Design, Distributed Systems, Test-Driven Development (pytest), Production Debugging & RCA, Data Structures & Algorithms

EDUCATION

University of Florida

Gainesville, FL

Bachelor of Science in Biomedical Engineering, GPA: 3.87

Aug. 2019 – May 2023